

1. 不同裝置上的結果

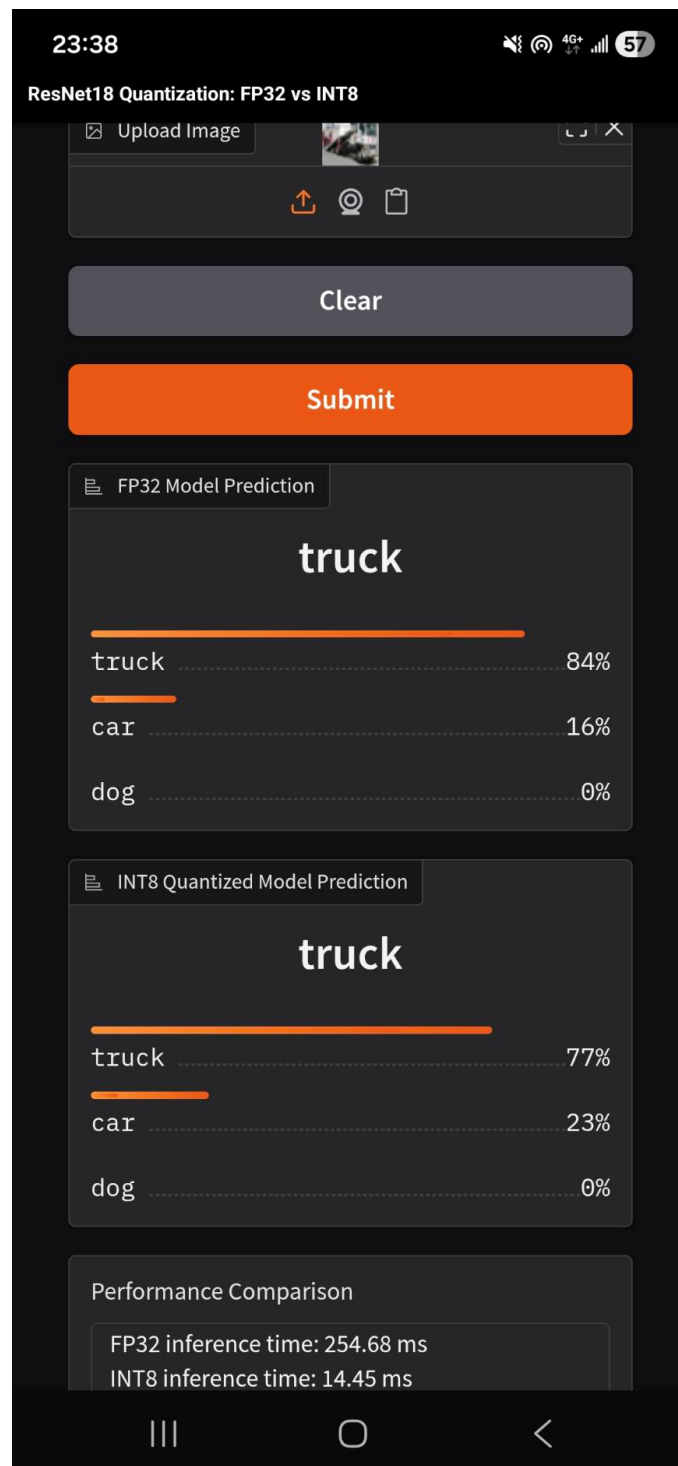
Device: Samsung A55 SmartPhone

Dataset: Cifar-10

File name: 10262.png

Class: truck

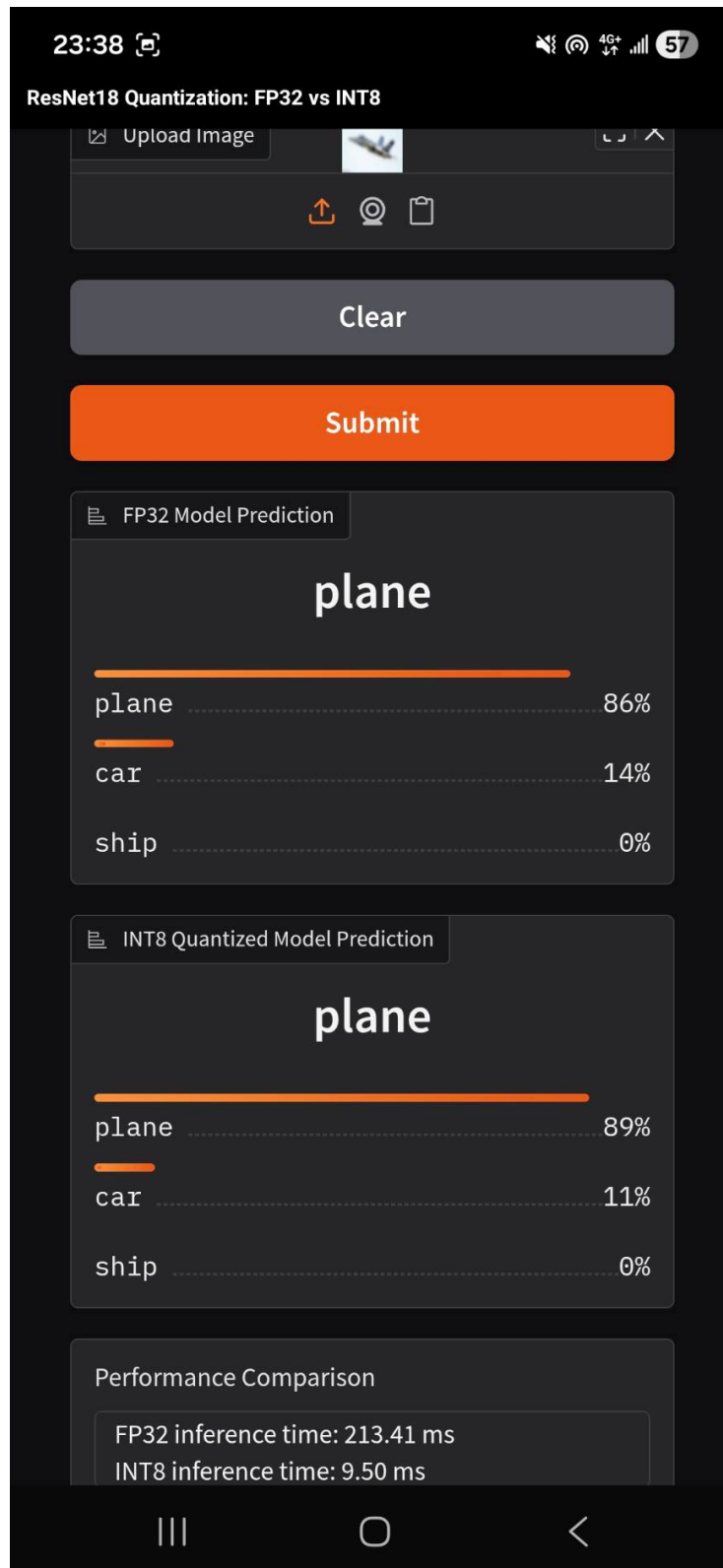
Result:



Filename: 10009.png

Class: plane

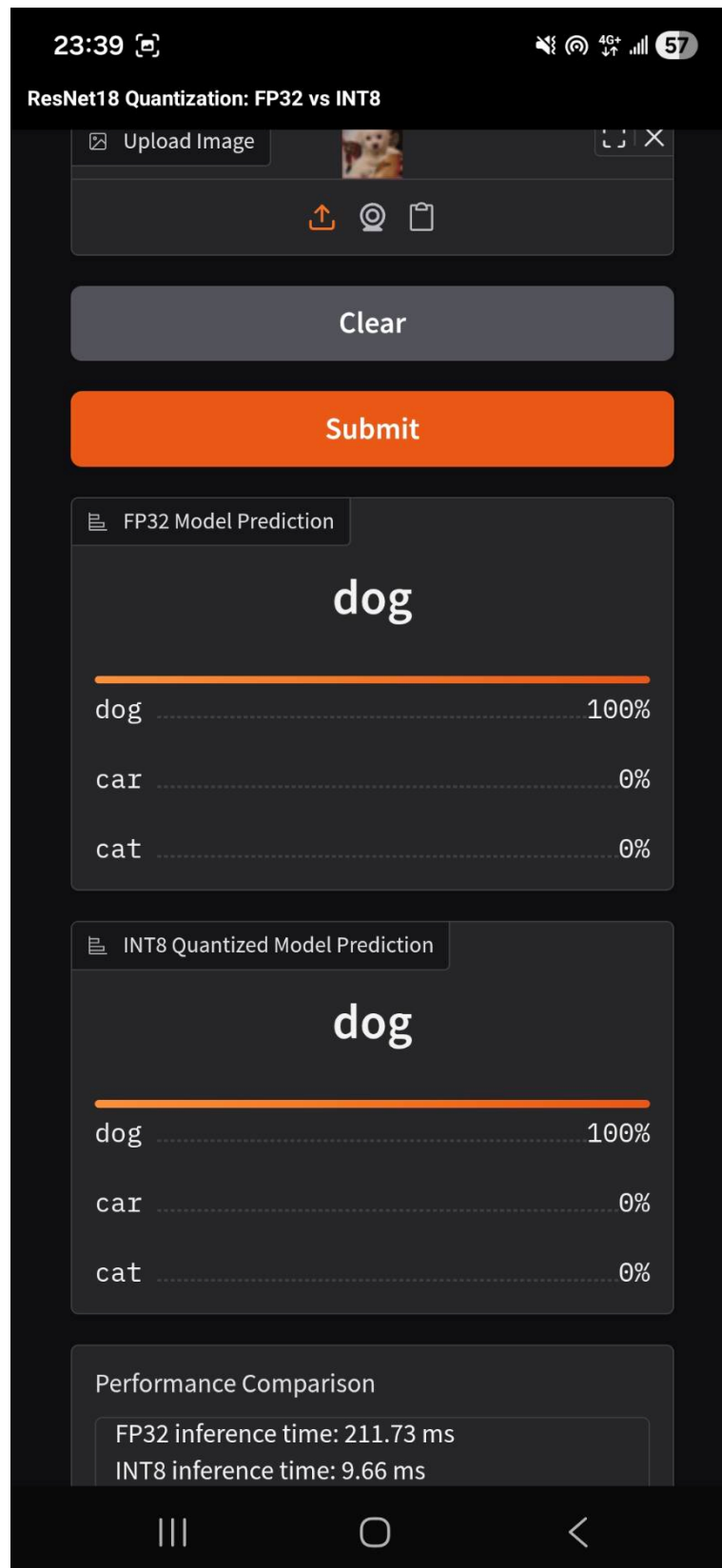
Result:



Filename: 10015.png

Class: dog

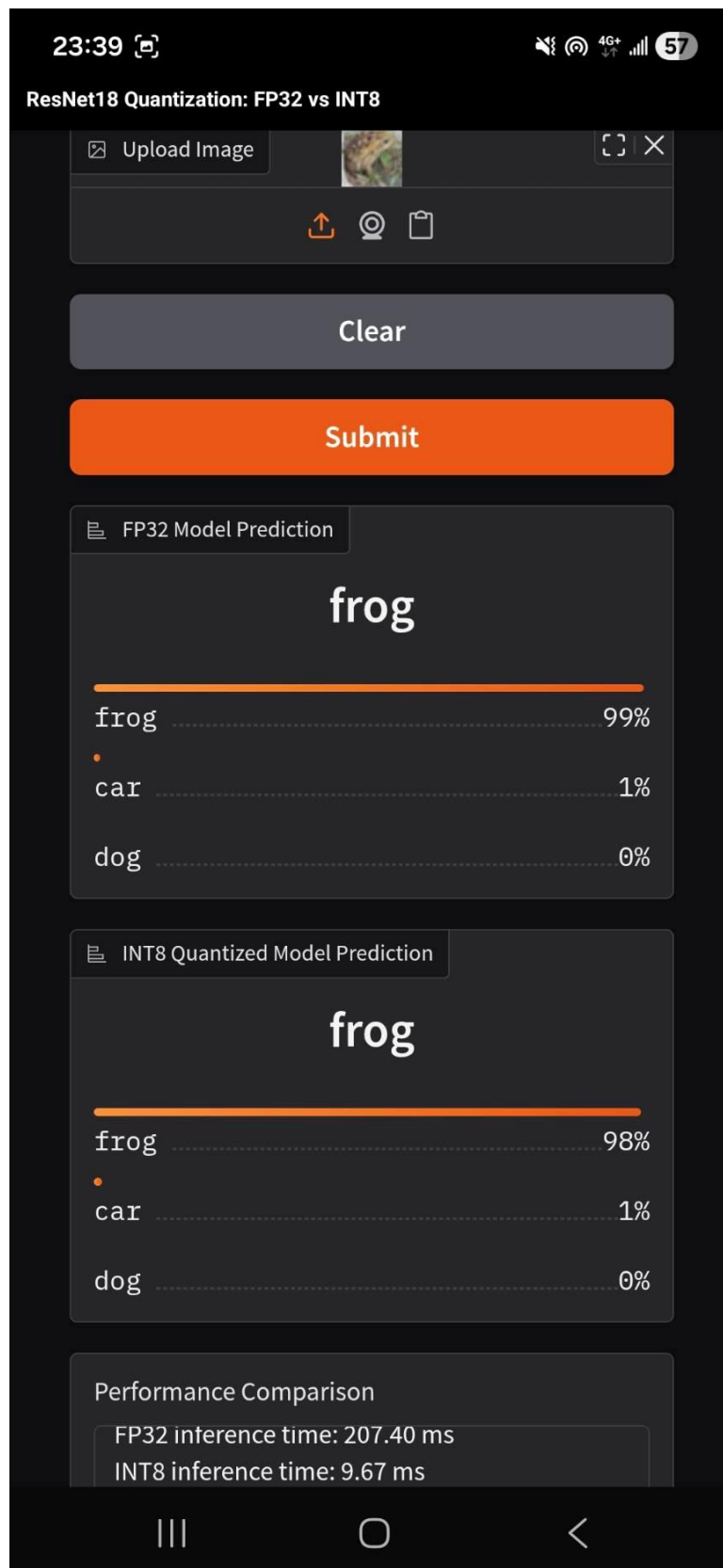
Result:



Filename: 9407.png

Class: frog

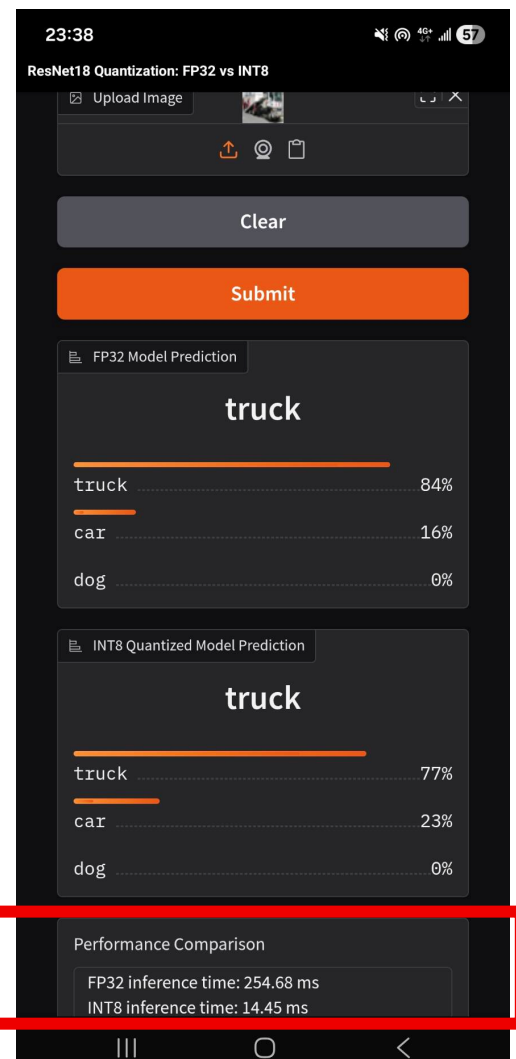
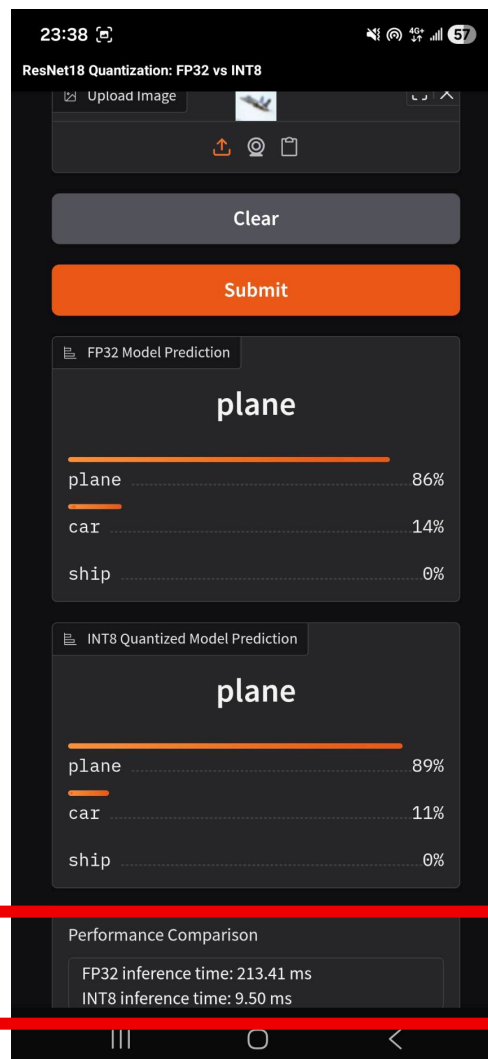
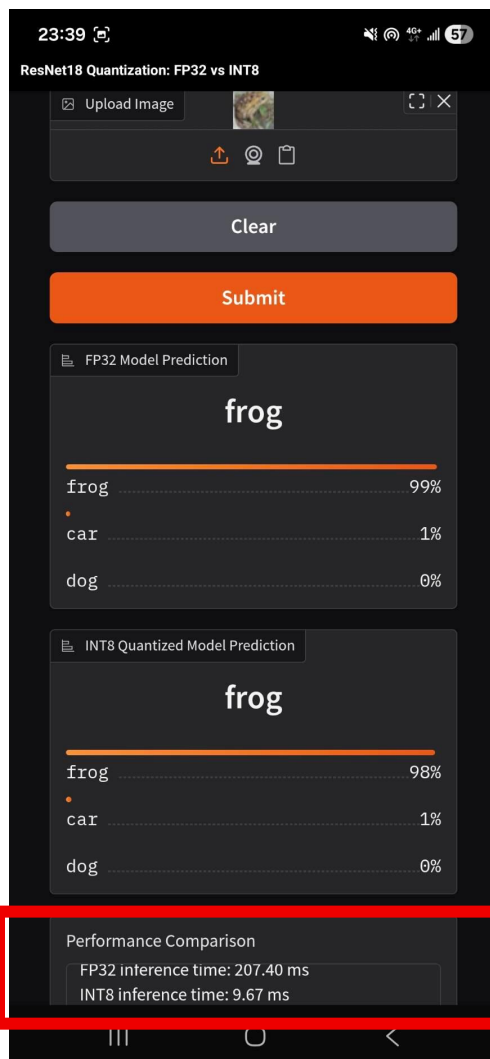
Result:



## 2. Compare\_fp32\_int8 function

從推論的時間可以得知，INT8 的推論比 FP32 來的快，FP32 和 INT8 的 預測結果大多數情況下應相同

```
... [Check] relative L2 diff FP32 vs INT8: 0.009550  
FP32 avg sec: 0.20993584156036377  
INT8 avg sec: 0.008017568588256837
```



在 code 的部分，主要針對 FP32 和 INT8 做推論過程的時間量測，比較時間以毫秒為單位，並且最後輸出在 UI 介面上會顯示兩個模型的 performance 的比較及 speedup 多少

```
# ===== Compare FP32 and INT8 =====
# TODO : Compare FP32 and INT8
def compare_fp32_int8(image: Image.Image):
    if image is None:
        return {}, {}, "Please Upload Your Image."
    if sess_fp32 is None:
        return {}, {}, (fp32_err or "The FP32 model has not been provided, so a comparison cannot be made.")

    x = preprocess(image)
    print(f"Preprocessed input shape: {x.shape}")
    # print(f"Preprocessed input data (first 5 elements): {x.flatten()[:5]}")

    # Your program
    # 取得 FP32 推論結果

    input_name_fp32 = sess_fp32.get_inputs()[0].name
    # Measure time
    start_time = time.time()
    out_fp32 = sess_fp32.run(None, {input_name_fp32: x})
    end_time = time.time()
    fp32_ms = (end_time - start_time) * 1000 # Convert to ms

    print(f"FP32 raw output (first 5 elements): {out_fp32[0][0][:5]}")
    p_fp32 = softmax_np(out_fp32[0][0])
    print(f"FP32 softmax probabilities (first 5 elements): {p_fp32[:5]}")
    print(f"FP32 softmax probabilities: {[LABELS[i], p_fp32[i]] for i in np.argsort(p_fp32)[-1][:3]}")
```

```
# 取得 INT8 推論結果 (假設 sess_int8 已建立)
if sess_int8 is not None:
    input_name_int8 = sess_int8.get_inputs()[0].name

    # Measure time
    start_time = time.time()
    out_int8 = sess_int8.run(None, {input_name_int8: x})
    end_time = time.time()
    int8_ms = (end_time - start_time) * 1000 # Convert to ms

    print(f"INT8 raw output (first 5 elements): {out_int8[0][0][:5]}")
    p_int8 = softmax_np(out_int8[0][0])
    print(f"INT8 softmax probabilities (first 5 elements): {p_int8[:5]}")
    print(f"INT8 softmax probabilities: {[LABELS[i], p_int8[i]] for i in np.argsort(p_int8)[-1][:3]}")

else:
    p_int8 = np.zeros_like(p_fp32) # 或處理錯誤
    int8_ms = 0

def top3_map(p):
    idx = np.argpartition(p, -3)[-3:]
    idx = idx[np.argsort(p[idx])[-1]]
    return {LABELS[i]: float(p[i]) for i in idx}

top3_fp32 = top3_map(p_fp32)
top3_int8 = top3_map(p_int8)

summary = (
    f"FP32 inference time: {fp32_ms:.2f} ms\n"
    f"INT8 inference time: {int8_ms:.2f} ms\n"
    f"Speedup (FP32/INT8): {(fp32_ms / max(int8_ms, 1e-9)):.2f}x"
)
return top3_fp32, top3_int8, summary
```

3. 在圖片輸入作為預測的部分，我認為可以提供幾個 sample 圖做為測試使用，雖然根據講義的要求明顯可猜到是要根據 model 輸入去找相關圖片是採用 cifar-10 dataset。